Minimizing Differences between Proofs and Final Production in Offset Printing

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Matching proofs on production presses is a typical problem ir Lithographic Offset Printing. Using frequency distributions, most common problem in matching was identified; using Ishikava cause and effect diagram the parameters to be examined were determined Experimental design was used to locate the best levels of parameters. The study resulted in reduction of non-conformance from 68% to 9%.

INTRODUCTION

The most common problem in printing multi-colour jobs on offset printing machines is in matching final printing- to the earlier printed proofs, approved by client.

The team after identifying the major factors effecting the problem, conducted experiments to determine the optimal operating conditions with respect to different factors. The paper describes the experiments and the results.

PROBLEM

Matching colours is a typical problem in lithographic offset printing. Colours in offset Printing can effectively be controlled by controlling two measurable characteristics-solid density and dot gain.

Based on long research & experience — Thomson Press. has made the specifications for solid density and dot gain. Study of past data through frequency distribution showed that production pressses are capable of printing within specified solid density and dot gain.

Where as mean dot gain level on proofing machines was significantly lower than the specified standards (values) — see Fig.1 (annexure). However performance regarding solid density was found to be satisfactory on proofing machines. Hence the problem was to increase the mean dot gain level on proofing machines to the specified standards.

THE PRODUCTION PROCESS

The major steps in printing of multi-colour jobs are

(a) Separation of four primary colours.

(b) Transferring image from colour separated positives to pre-sensitized metallic plates, thus creating image and non-image area on the same plane of metallic plate by photo lithographic process.

(c) Printing multi-colour images on paper by transferring four basic primary colours subsequently one over another.

THE WORK PLAN OF THE TEAM

The team investigated the following

- The existing level of dot gain on proofing machines.
- The factors effecting dot gain.
- The parameters effecting the dot gain.
- Effect of the change in levels of parameters.

The team established for itself the following schedule of activities —

- Study dot gain for one month under existing process parameters.
- Classification of factors effecting dot gain - men, machine, material and method.
- Conducting designed experiments to solve the problem based on the analysis of the possible factors causing dot gain.
- Determining optimal levels, through analysis of experimental data.
- Implementation.

THE ISHIKAWA FISH BONE DIAGRAM

Once the core of problem was known, the team sat down and suggested the possible causes of the problem. All the suggestions were then arranged into fine categories — Positives, plates, paper, ink and machines. Some of the most significant of these causes are listed below and presented as an Ishikawa Fish Bone diagram in Fig 2 (annexure).

- Positives : Dot Size
- Plate : Exposure time
- Paper : Roughness
- Ink : Amount of Ink
- Machine : Impression pressure & Temperature

THE EXPERIMENTAL DESIGN & RESULTS

From all these factors, four were chosen as the most dominating ones. These factors and the levels chosen are —

A : Impression pressure
  A1 : 0.90 mm A2: 0.85 mm
B : Tack of ink
  B1 : Low B2 : High
C : Plate Exposure time
  C1 : Normal C2 : Half
D : Plate bed temperature
  D1 : 16°C D2 : 18°C

Since each of the factors has two levels, a full factorial design would have required 2^4 (i.e. 16) combinations, which were considered to be too many. Further on technical arguments it was concluded that only main effects of A, B, C & D and two factor interections — AB, AD & BD are of interest.

These factors were assigned to linear graph as shown in Fig 3.

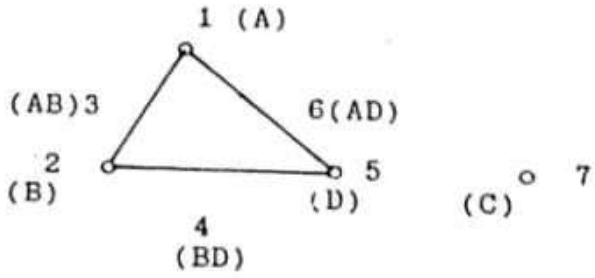

Fig. 3. Linear Graph

A standard test forme was produced by the process department. The test forme was then printed on proofing machines on paper (Glazed news print) under the treatment combinations defined by the Orthogonal Array as shown in Table 1.

Table 1 : The layout of the experiment

S.No.	Factors & their levels			
	A	B	C	D
1.	.90mm	Low	Half	16°C
2.	.90mm	Low	Normal	18°C
3.	.90mm	High	Normal	16°C
4.	.90mm	High	Half	18°C
5.	.85mm	Low	Normal	16°C
6.	.85mm	Low	Half	18°C
7.	.85mm	High	Half	16°C
8.	.85mm	High	Normal	18°C

mm : milli meter

Based on experimental data, the main effects & interactions for significant factors were estimated. The effect of factor is defined to be the change in response produced by a changed in the level of the factor. Presence of interactions between the factors means that the difference in response between the levels of one factor is not the same at all levels of the other factor. The main effects and interactions are illustrated in Fig 4.

Using the estimates of significant main effects and interactions, expected dot gains under each of the $2^4 = 16$ treatment combinations (4-factors, each at 2-levels) were computed and are given in Table 2.

Table 2 : Expected dot gain under different treatment combinations

Treatment Combination	Expected Dot Gain
A1B1C1D1	16.33
A2B1C1D1	20.16
A1B2C1D1	10.08
A2B2C1D1	13.91
A1B1C2D1	9.75
A2B1C2D1	13.58
A1B2C2D1	3.50
A2B2C2D1	7.33
A1B1C1D2	18.66
A2B1C1D2	14.83
A1B2C1D2	15.41
A2B2C1D2	11.58
A1B1C2D2	12.08
A2B1C2D2	8.25
A1B2C2D2	8.83
A2B2C2D2	5.00

Standard Deviation σ = 1.75
Specifications = 20 ± 3

RECOMMENDATIONS & CONCLUSIONS

On the basis of above investigations, the optimal levels for the parameters were established and the expected performance under them were estimated. These are given in Table 3.

Table 3 : Optimal process parameters for proofing machines (FAG) and the expected performance

Factors	Recommended Level
Impression	0.85 mm
Tack of Ink	Low
Plate exposure time	Half
Plate bed temperature	16°C
Expected Performance	
Expected dot gain	= 20.16
Standard deviation	= 1.75
Expected % conformance	= 91.20

The recommendations were implemented and the results obtained are given in Table 4.

Table 4 : Results of dot gain during trial implementations

Colour	$\hat{\mu}$	$\bar{R}$	$\hat{\sigma}$	P	P'
Black	19.6	2.0	1.8	90.2	32.4
Cyan	18.6	1.0	0.9	98.3	22.1
Magenta	20.6	1.6	1.4	94.8	12.7
Yellow	21.6	1.1	1.0	92.0	17.1

μ : Mean ; R: Range ; σ: R/d_2
P : % Conformace after study
P': % Conformance before study

The systematic analysis of the problem, design & analysis of experiments and implementations of the results, led directly to a significant savings due to —

(a) reduction of reprints for rejected proofs
(b) the time saved on machine trying to match the proofs not produced to meet the press conditions

The entire project was carried out over a three months period. So the expected savings are supposed to be many times than the investment in the project.

Acknowledgements
We are thankful to Prof S.C. Chakravorty, who reviewed the study periodically in detail and made a number of incisive recommendations and several improvements in it.

ANNEXURE

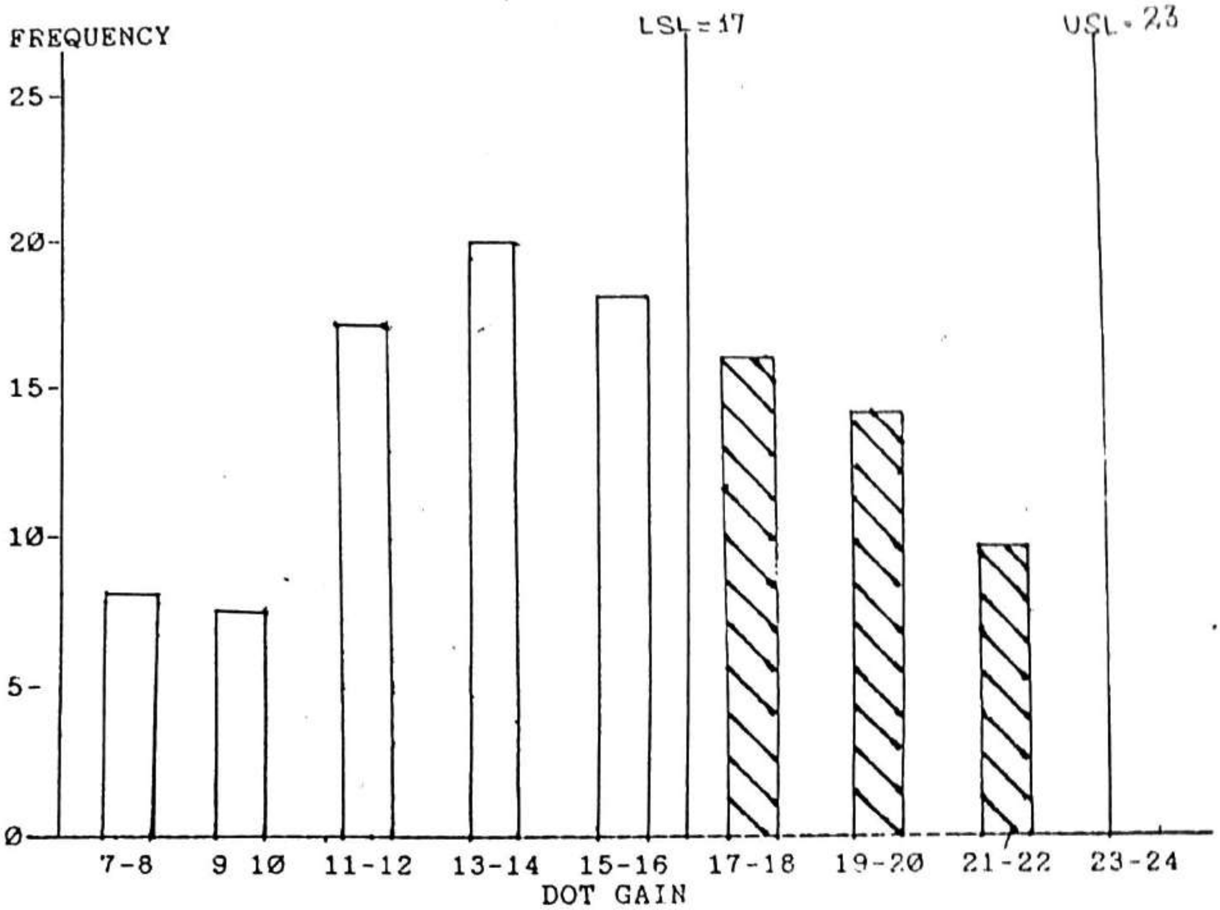

Fig. 1: Frequency distribution for dot gain on proofing machines during October'88.

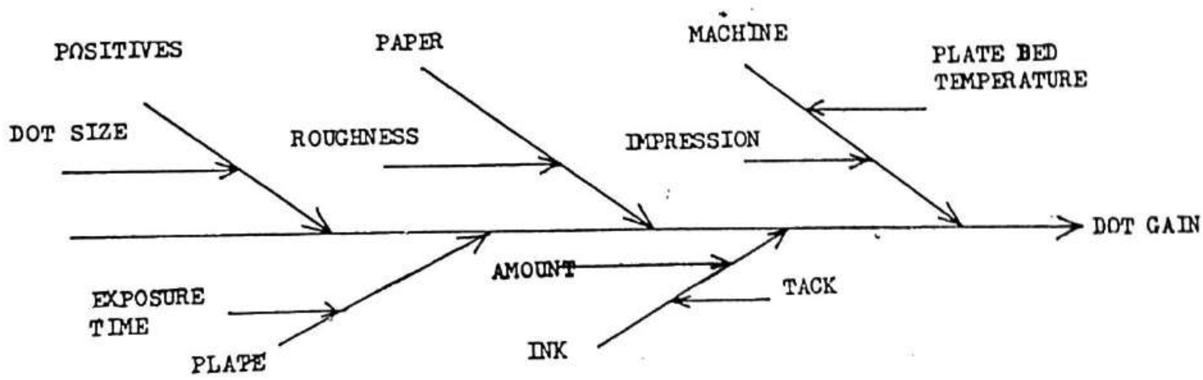

Fig. 2: The Ishikawa Fish Bone Diagram

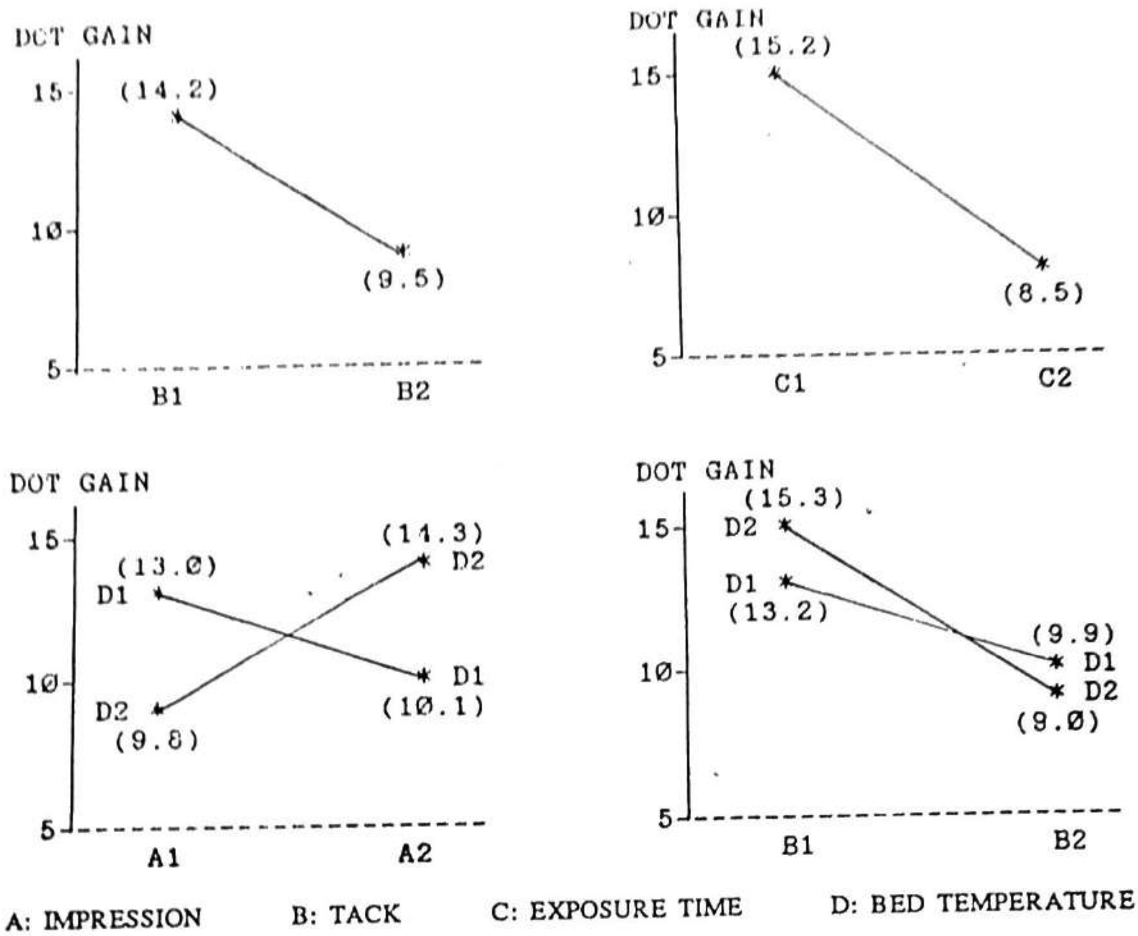

Fig. 4: Plot for significant main effects and interections